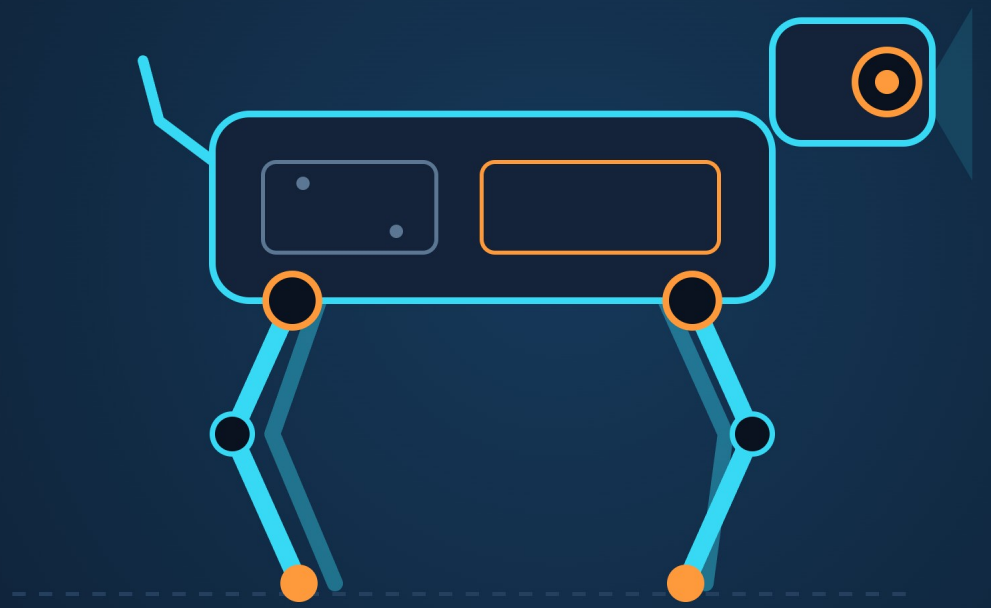
OPEN-SOURCE QUADRUPED // RASPBERRY PI 3B

SPOT MICRO

AI Quadruped

A robot dog that hears offline and understands the world through a vision-language model.



OFFLINE VOICE

VISION + LLM

12 SERVOS

What we are building

The concept

An intelligent, autonomous quadruped robot based on the open-source Spot Micro platform.

OPEN SOURCE BASE

Core goal

One versatile assistant for home aid, emotional companionship and active security inspection.

HOME / COMPANION / SECURITY

Foundation

Low-level inverse kinematics and motion control on custom 3D-printed mechanics, with vision-based spatial awareness.

SIT / STAND / TILT / WALK

Motivation and relevance

VERSATILE PHYSICALITY

Legs, not wheels

Four legs move far better on complex surfaces than wheels, a natural fit for real-world environments.

FUNCTIONAL AI INTEGRATION

Beyond a demo

Practical work such as autonomous patrolling and surveillance, matching the roles of traditional service dogs.

HYBRID INTELLIGENCE

Hybrid brain

Fast offline voice with Vosk plus multimodal LLMs (Gemini / Claude): responsive and context-aware, with onboard security kept intact.

Offline ears. Cloud brain.

ON THE ROBOT

Vosk speech-to-text

Low-latency recognition that runs on the Raspberry Pi, with no network needed to hear a command.

EDGE-CLOUD

Multimodal LLM

Gemini or Claude reads the camera frame together with the request and plans what the robot does next.

Key capabilities

01 // LOCOMOTION

Gait and kinematics

Precise locomotion, stance changes and balance across varied terrain.

02 // INTERACTION

Voice and vision

Live speech recognition combined with camera-feed analysis for natural interaction.

03 // NAVIGATION, PLANNED ADD-ON

Spatial navigation

Collision avoidance and room mapping with RPLidar SLAM.

04 // COMPANION

AI companion

Understands complex context, carries out directives and works as a daily assistant.

What we build it from

COMPUTE

Raspberry Pi 3B

240 GB SSD over USB-SATA

Pi Camera Module v2

MOTION

12 PDI-HV5523MG servos

PCA9685, 16-channel PWM over I²C

CHASSIS

Custom 3D-printed Spot Micro frame

Mounts and a body plate redesigned for the Pi

POWER

2S LiPo, 7.4 V, 4000 mAh

5 V / 5 A UBEC for the Pi and PCA9685

VOICE AND AUDIO

USB omnidirectional mic

AUX or USB mini speaker

LIDAR, PLANNED ADD-ON

RPLidar A1

SLAM and room mapping

From voice to motion

INPUT
Microphone
USB omni mic

→

OFFLINE
Vosk STT
Runs on the Pi

→

LOGIC
Intent router
Command or chat

→

EDGE-CLOUD
Gemini / Claude
Text plus Camera v2 frame

↓

ACTUATORS
12 servos
PDI-HV5523MG

←

I²C PWM
PCA9685
16-channel driver

←

KINEMATICS
Gait and IK
Joint angles for 4 legs

←

OUTPUT
Action plan
Move, pose or spoken reply

Preparation and modelling

DONE	Concept and scope: open-source Spot Micro platform
DONE	Parts list: compute, servos, power, camera and audio
DONE	Body plate remodelled for Raspberry Pi 3B: 4 × M2.5 holes, 58 × 49 mm
READY	Software architecture: offline STT, VLM planning, gait and IK control
NEXT	Print the parts, assemble the legs and electronics
PLANNED	Bring-up: kinematics, camera, voice and VLM integration
PLANNED	RPLidar A1 and SLAM as a later upgrade

Build plan in four phases

01
Build
Print the frame, mount the electronics and wire the power.

02
Move
Calibrate servos, then stand, sit, tilt the body and walk.

03
Talk and see
Add Vosk voice commands, the camera stream and model replies.

04
Explore
Optional upgrade: RPLidar A1, SLAM and patrol routes.

SPOT MICRO // AI QUADRUPED

Thank you

Questions and ideas